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MILITARY STANDARD

FASTENER TEST METHODS

METHOD 28,

ELEVATED TEMPERATURE DOUBLE SHEAR

FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Fastener Test Methods, Method 28, Elevated Temperature Double Shear

MIL-STD-1312-28

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FOREWORD

This standard sets forth a standard test procedure to determine double shear strength, at elevated temperatures, for all types of structural fasteners.

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1. SCOPE

1.1 Applicability. This method outlines a standard procedure for testing all types of structural fasteners in double shear at elevated temperatures. This procedure is intended to define the test required to determine the double shear strength of the fastener itself, not the strength of the fastener in any material combinations.

1.2 For purpose of this test method, elevated temperatures shall be in the following classes:

Class	Temperature Range
I	250°F through 700°F
II	701°F through 1200°F
III	1201°F through 1800°F

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662 Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 Surface Texture (Surface Roughness, Waviness and Lay)

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 4 Load Verification of Testing Machines

ASTM (Proposed) Preparation of Thermocouple Measuring Junctions

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of applying a compressive load at a controllable rate. The calibration system for the machine shall conform to MIL-STD-45662. Its accuracy shall be verified every 6 months by a method complying with ASTM E 4, using a calibration device which shall have been calibrated by the National Bureau of Standards not more than 2 years before its use. The loads of the fasteners tested shall be within the loading range of the testing machine as defined in ASTM E 4.

4.1.2 Test fixture. Fixtures shall be fabricated from materials capable of withstanding the test loads without permanent deformation at the test temperatures, except for slight wear of the block and blade shear faces. Test fixtures shall be in accordance with figure 1.

4.1.3 Controls. Controls and thermocouples used to indicate and control the temperature within the furnace shall be capable of determining, indicating and controlling the temperature within $\pm 4^{\circ}$F for Class I and within $\pm 5^{\circ}$F for Classes II and III.

4.1.3.1 Thermocouples. Elevated test temperature must be sensed by thermocouples made from selected and calibrated wire certified by the producer. Wire gage is dictated by the temperature range and service life. However, the smallest diameter possible should be used to obtain reliable measurement in each range. Thermocouples shall not be used more than 10 times. They should have any heat affected zone removed and be re-beaded before reuse. Thermocouples shall be installed on the test fastener so as to indicate the temperature of the fastener and to exclude indications of the radiant heat from the furnace walls. Thermocouples shall meet ASTM (Proposed), Preparation of Thermocouple Measuring Junctions.

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4.1.3.2 Thermocouple attachment and location. Thermocouples shall be mechanically attached at the top and bottom of the fastener to maintain a constant record of temperature level and gradient throughout the testing period.

4.1.3.3 Gradient. The temperature gradient of fasteners under test shall be not greater than 10°F.

4.1.4 Shear preload. The test apparatus shall not exert a shearing load during the time the specimen is being brought up to test temperature.

4.2 Test specimens.

4.2.1 General. When fasteners which are supplied complete in one assembly are being evaluated, the tests shall be performed on the entire fastener assembly. Tests resulting in failure of any component of the fastener assembly shall be considered a test of the fastener.

4.2.2 Grip length. Unless specified otherwise, protruding head fasteners to be tested shall have a grip length of at least two times the fastener shank diameter. For flush head fasteners, a grip length of at least three times the fastener's shank diameter shall apply. The test shall be performed on the shank portion of the fastener.

4.2.3 Preload. The fastener shall not exert tension preload or clamping force on the test block. Preload may be released by the melting of a Cerrolow spacer or equivalent.

5. DETAIL REQUIREMENTS

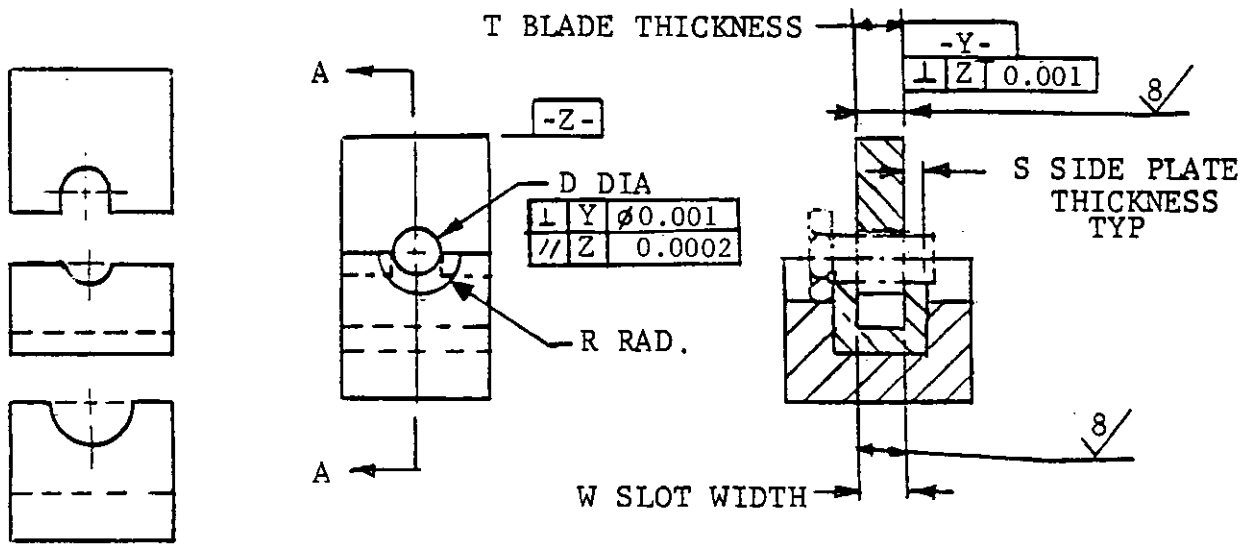
5.1 Test procedures.

5.1.1 Test setup.

5.1.1.1 Assembly. The fastener shall be installed in the shear block as specified in figure 1. The shear blade portion shall be placed in position on the fastener. The assembly of fastener and fixtures shall be placed between the heads of the loading machine. The load shall be applied perpendicular to the axis of the fastener within one degree (1°).

5.1.2 Loading rate. The rate of application of load shall be not greater than 100,000 pounds per square inch of nominal double shear area per minute with a tolerance of $\pm$10% (see table I).

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NOMINAL FASTENER SIZE	D		S +0.0010 -0.000	T +0.000 -0.0010	W +0.0020 -0.0010	R REF
	MAX	MIN				
0.125	0.1260	0.1253	0.063	0.1250	0.1250	0.156
0.156	0.1573	0.1566	0.078	0.1563	0.1563	0.190
0.164	0.1650	0.1643	0.083	0.1562	0.1562	0.190
0.190	0.1885	0.1878	0.094	0.1875	0.1875	0.219
#10	0.1910	0.1903	0.094	0.1900	0.1900	0.219
0.250	0.2510	0.2503	0.125	0.2500	0.2500	0.281
0.313	0.3135	0.3128	0.156	0.3125	0.3125	0.344
0.375	0.3760	0.3753	0.188	0.3750	0.3750	0.406
0.438	0.4385	0.4378	0.219	0.4375	0.4375	0.469
0.500	0.5010	0.5003	0.250	0.5000	0.5000	0.531
0.563	0.5635	0.5628	0.281	0.5625	0.5625	0.594
0.625	0.6260	0.6253	0.312	0.6250	0.6250	0.656
0.750	0.7510	0.7503	0.375	0.7500	0.7500	0.781
0.875	0.8760	0.8753	0.437	0.8750	0.8750	0.906
1.000	1.0010	1.0003	0.500	1.0000	1.0000	1.031
1-1/8	1.1260	1.1253	0.562	1.1250	1.1250	1.156
1-1/4	1.2510	1.2503	0.625	1.2500	1.2500	1.281
1-3/8	1.3760	1.3753	0.687	1.3750	1.3750	1.406
1-1/2	1.5010	1.5003	0.750	1.5000	1.5000	1.531

NOTES:
1. Break shear edges minimum 0.005. Jig shall be reworked when wear results in chamfer or radius of 0.010.
2. For fixture sizes other than those listed, the geometric proportions specified herein shall be maintained.
3. Dimensions in inches.

FIGURE 1. Double shear fixture.

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TABLE I. Double shear load rates.

NOMINAL DIAMETER	LOAD RATE LB/MIN	NOMINAL DIAMETER	LOAD RATE LB/MIN
0.125	2,480	0.563	49,600
0.156	3,840	0.625	61,200
0.164	4,200	0.750	88,000
0.190	5,600	0.875	120,000
0.250	10,000	1.000	156,000
0.313	15,400	1.125	200,000
0.375	22,000	1.250	244,000
0.438	30,000	1.375	296,000
0.500	39,200	1.500	352,000

The testing laboratory, at their option, may use a constant strain rate which will produce the specified load rate (±10 percent) in the elastic range. That is, the strain rate shall be equal to the initial stress rate of 50,000 pounds per minute per square inch divided by the elastic modulus in shear.

CAUTION

When loading, ensure that lag between actual and indicator has been taken into account.

5.1.3 Ultimate load. The ultimate load is the highest load recorded. The test may be discontinued and the fastener removed without a complete shear failure, after the ultimate load has been reached.

5.1.4 Temperature. The maximum time for heating the fastener to temperature shall be specified in the procurement document. The time shall not be less than 60 minutes per inch of fastener diameter. Unless otherwise specified, the test setup shall be maintained at the test temperature 30 minutes before load application and throughout the test load application. Do not heat the fastener to a temperature higher than the test temperature.

6. NOTES

6.1 Test report. The test report shall contain the following data:

a. Fastener Descriptions.

1. Part number.
2. Lot identification.
3. Manufacturer.

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4. Fastener material.
5. Heat treat.
6. Measured fastener diameter.
7. Grip length.

b. Test Machine.

1. Model and Serial Number.
2. Calibration Date
3. Verification Date.

c. Ultimate Load.

d. Fixture Description, Material and Photograph.

e. Ultimate Load and Load Rate.

f. Temperature Readings.

g. Type of Heat Used.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Navy - AS, SH
Air Force - 99
DLA - IS

Preparing activity:
Navy - AS
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